Lexicon of Formulae

Classic Squared Identities

- $(a+b)^2 = a^2+b^2+2ab$
- $(a-b)^2 = a^2+b^2-2ab$
- $(a^2+b^2) = (a+b)^2-2ab$
- $(a^2-b^2) = (a+b)(a-b)$
- $(x+a)(x+b) = x^2+x(a+b)+ab$
- $(x-a)(x-b) = x^2-x(a+b)+ab$
- $(ax+b)(cx+d) = acx^2 + (ad + bc)x + bd$

Triple Squared Identities

- $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$
- $(a+b-c)^2 = a^2+b^2+c^2+2ab-2bc-2ca$
- $(a^2+b^2+c^2-ab-bc-ca) = ½\{(a-b)^2+(b-c)^2+(c-a)^2\}$
- $(a^2+b^2+c^2-ab-bc-ac \geq 0)$
- If $a \neq b \neq c$ then $a^2+b^2+c^2-ab-bc-ac>0$

Classic Cubic Identities

- $(a+b)^3 = a^3+3ab(a+b)+b^3 => a^3+b^3+3a^2b+3ab^2$
- $(a-b)^3 = a^3-3ab(a-b)-b^3 => a^3-b^3-3a^2b-3ab^2$
- $(a^3+b^3) = (a+b)(a^2-ab+b^2)$
- $(a^3-b^3) = (a-b)(a^2+ab+b^2)$

Triple Cubic Identities

- $(a^3+b^3+c^3-3abc) = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
- If $a+b+c=0$, $(a^3+b^3+c^3)=3abc$
- $(a+b+c)^3 = a^3+b^3+c^3+3(a+b)(b+c)(c+a)$

x+1/x Identities

- If $(x+1/x)=a$, $(x^2+1/x^2) = a^2-2$
- If $(x-1/x)=a$, $(x^2+1/x^2) = a^2+2$
- If $(x+1/x)=a$, $(x^3+1/x^3) = a^3-3a$
- If $(x-1/x)=a$, $(x^3-1/x^3) = a^3+3a$
- $(x^n+1/x^n)=(x+1/x)(x^{n-1}+1/x^{n-1})-(x^{n-2}+1/x^{n-2})$

Sophie-Germain's Identity

$x^4+4y^4 = \{(x+y)^2+y^2\}\{(x-y)^2+y^2\}$

Alternative:

$x^4+4y^4 = (x^2+2xy+2y^2)(x^2-2xy+2y^2)$

Pythagorean Theorem & Triplets

- Theorem: $a^2+b^2=c^2$
- Triplets: (for $m>1$)
    - $2m$ [even]
    - (m^2-1) [orthogonal]
    - (m^2+1) [greatest]
- Triplet Generator: $(2m)^2+(m^2-1)^2=(m^2+1)^2$

Areas of Shapes

- Square: s^2 [*s* represents side]
- Rectangle: $l*b$ [*l* represents length & *b* represents breadth]
- Circle: πr^2
- Rhombus: $½*d_1*d_2$ [d_1 & d_2 are individual diagonals]
- Trapezium: $½*(a+b)*h$ [a & b are parallel sides while h represents height]
- Parallelogram: $b*h$ [b represents base & h represents height]
- Triangle: $½*b*h$ [b represents base & h represents height]
- Equilateral Triangle: $((\sqrt{3})/4)*a^2$ [*a* represents side]

- Heron's Formula: $\sqrt{(s(s-a)(s-b)(s-c))}$ [a, b, c are side lengths while s represents semi-perimeter $(a+b+c)/2$]
- Semi-perimeter of a Triangle $= (a+b+c)/2$
- General Quadrilateral: $½*d*(h_1+h_2)$ [d represents diagonal while h_1 & h_2 are individual heights]
- Cyclic Quadrilateral (Brahmagupta's Formula): $\sqrt{((s-a)(s-b)(s-c)(s-d))}$ [a, b, c, d are side lengths while s represents semi-perimeter $(a+b+c+d)/2$]
- Semi-perimeter of a Quadrilateral $= (a+b+c+d)/2$
- Sector of a Circle: $(\theta/360) * \pi r^2$
- Minor Segment of a Circle: Area of a Sector – Area of Triangle Formed
- Major Segment of a Circle: Area of a Circle – Area of a Minor Segment
- Quadrant: $(\pi r^2)/4$

<u>Perimeter/Circumference of Shapes</u>

- Square: $4s$
- Rectangle: $2(l+b)$
- Circle: $2\pi r$
- Rhombus: $4s$
- Trapezium: $a+b+c+d$
- Parallelogram: $2(a+b)$
- Isosceles/Scalene Triangle: $a+b+c$
- Equilateral Triangle: $3s$
- Sector of a Circle: $2r+((\theta/360)*2\pi r)$
- Arc of a Circle: $(\theta/360)*2\pi r$
- Semicircle: πr
- Quarter-circle: $(\pi r)/2$

<u>Pi (π)</u>

$\pi =$ Circumference / Diameter

Basic Algebraic-Geometric Equalities of Polygons

Here, *n* represents the number of sides of the polygon.

- Number of Diagonals: $n(n-3)/2$
- Sum of interior angles: $(n-2)*180°$
- Individual interior angles (for regular polygons): $((n-2)*180°)/n$
- Sum of exterior angles: $360°$
- Individual exterior angles: $360°/n$

Surface Area of Polyhedrons

- TSA of Cube: $6a^2$
- TSA of Cuboid: $2(lb+bh+lh)$
- TSA of Cylinder: $2\pi r(h+r)$
- TSA of Cone: $\pi r(l+r)$
- TSA of Hemisphere: $3\pi r^2$
- LSA of Cube: $4a^2$
- LSA of Cuboid: $2(l+b)*h$
- CSA of Cylinder: $2\pi rh$
- CSA of Cone: πrl
- SA of Sphere: $4\pi r^2$
- CSA of Hemisphere: $2\pi r^2$
- TSA of Hemisphere: $3\pi r^2$
- Cone Slant Height: $l = \sqrt{(r^2+h^2)}$
- CSA of Hollow Hemisphere: $2\pi(R^2+r^2)$
- TSA of Hollow Hemisphere: $\pi(3R^2+r^2)$
- For hollow hemisphere, *R* represents radius of bigger hemisphere while r represents radius of smaller hemisphere.
- CSA of Frustum of a Cone: $\pi l(r_1+ r_2)$
- TSA of Frustum of a Cone: $\pi l(r_1+ r_2) + \pi l((r_1)^2+ (r_2)^2)$
- Length of a Frustum, derived from Pythagores Theorem: $\sqrt{(h^2+(r_2-r_1)^2)}$

- LSA of Triangular Prism: l * base perimeter
- TSA of Triangular Prism: (l * base perimeter) + 2 * area of base
- Base Perimeter of Triangular Prism (sum of edges of triangular face): a+b+c
- LSA of Triangular Pyramid: Area of side faces
- TSA of Triangular Pyramid: Area of side faces + Area of base

Volume of Polyhedrons

- Cube: a^3
- Cuboid: l*b*h
- Cylinder: $\pi r^2 h$
- Hollow Cylinder: $\pi(R^2)h - \pi(r^2)h$ [Here, *R* represents radius of bigger cylinder while r represents radius of smaller cylinder]
- Cone: $(⅓)\pi r^2 h$
- Sphere: $(4/3)\pi r^3$
- Hemisphere: $(⅔)\pi r^3$
- Hollow Hemisphere: $(⅔)\pi(R^3-r^3)$ [Here, *R* represents radius of bigger hemisphere while r represents radius of smaller hemisphere]
- Frustum of a Cone: $(⅓)\pi h((r_1)^2+ (r_2)^2+ (r_1)(r_2))$
- Triangular Prism: Area of Base * h
- Triangular Pyramid: ⅓ * Area of Base * h

Simple Interest (S.I.)

P represents principal, *R* represents rate, *T* represents time.

S.I. = (PRT)/100

A = P+S.I.

Compound Interest (C.I.)

P represents principal, *r* represents rate, *n* represents time [quarterly/half-yearly/annually], *A* represents amount.

$A = P(1+[r/100])^n$

C.I. = A-P

- For *half-yearly*, r/2 and 2n
- For *quarterly*, r/4 and 4n

Profit, Loss, CP and SP

- Profit = SP-CP (if SP>CP)
- Loss = CP-SP (if CP>SP)
- P% = (Profit/CP)*100
- L% = (Loss/CP)*100
- P% = (Increase/Original)*100
- L% = (Decrease/Original)*100
- SP = ((100+P%)/100)*CP
- CP = (100/(100+P%))*SP
- SP = ((100-L%)/100)*CP
- CP = (100/(100-L%))*SP

MP and Discount

- SP = MP - Discount
- D% = (D/MP)*100
- SP = ((100-D%)/100)*MP
- SP = $((100-D_1)/100)*((100-D_2)/100)*...*MP$ *(successive discount)*

Coordinate Geometry

- Distance Formula (between two points): $\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2]}$
- Midpoint Formula (between two points): $((x_1+x_2)/2, (y_1+y_2)/2)$
- Section Formula (trisection): $((m_1x_2+m_2x_1)/(m_1+m_2), (m_1y_2+m_2y_1)/(m_1+m_2))$
- Area of a Triangle with vertices: $½*|(x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2))|$
- Collinearity of three points (Collinearity Checker): AB+BC=CA

Linear Formulae

- Standard form of Linear equation: $ax+by=C$
- Linear Relationship: $y = ax+b$ (where *a* is the slope, and *b* is the y-intercept)

Quadratic Formulae

Standard form of Quadratic equation: $ax^2+bx+c=0$ (where $a\neq0$)

- Quadratic Formula: $(-b\pm\sqrt{(b^2-4ac)})/2a$
- Discriminant of Quadratic Formula: $D = b^2-4ac$

Real and Imaginary values based on State of Discriminant:

- If $D>0$: Two different real roots
- If $D=0$: Two equal real roots
- If $D<0$: No real roots (all imaginary roots)

For two roots α *and* β*, the relation is:*

- $(\alpha+\beta) = -(b/a)$
- $(\alpha*\beta) = (c/a)$

Cubic Formulae

- Standard form of Cubic equation: $ax^3+bx^2+cx+d=0$ (where $a\neq0$)

Probability

- Experimental Probability of an event E: (number of trials)/(total number of trials)

Proportionality

- Basic Proportion (Ratio): $a/b = c/d$
- Mean Proportional (in $a:b = b:c$): $b = \sqrt{(a*c)}$
- Third Proportional: $c = b^2/a$
- Fourth Proportional: $d = (b*c)/a$
- Direct Proportion: $y = kx$ ($y \propto x$)
- Inverse Proportion: $y = k/x$ ($y \propto 1/x$)
- *k* represents the proportionality constant.

Interesting Patterns

- $(n-1)^2 + (n+1)^2 - 2n^2 = 2$

Facts

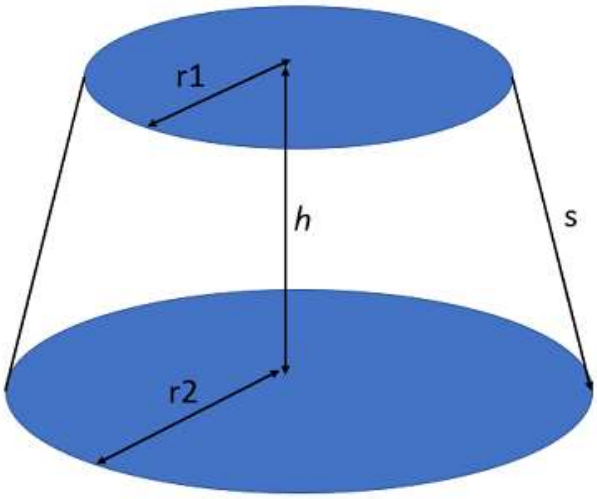
Geometry

- Diameter of a circle is its longest chord.
- π (Pi) represents the ratio between circumference and diameter of a circle. It is a constant that holds true for all circles and is an irrational value, meaning it is non-terminating and cannot be represented as a fraction.
- θ (Theta) usually represents an angle.
- The sum of the internal angles of any triangle is always 180° degrees.
- The sum of the internal angles of any quadrilateral is always 360° degrees.
- The sum of the exterior angles of any polygon is always 360° degrees.

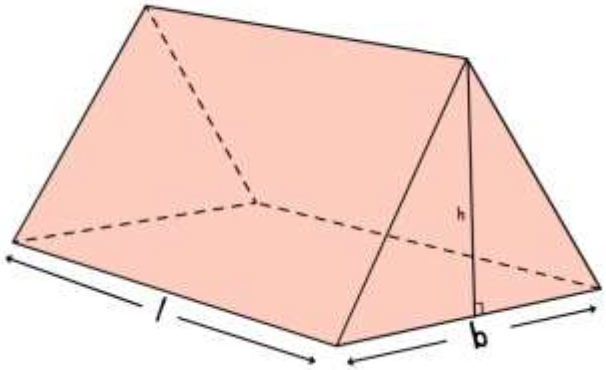
Probability

- Each possible outcome is called an element of the sample space.

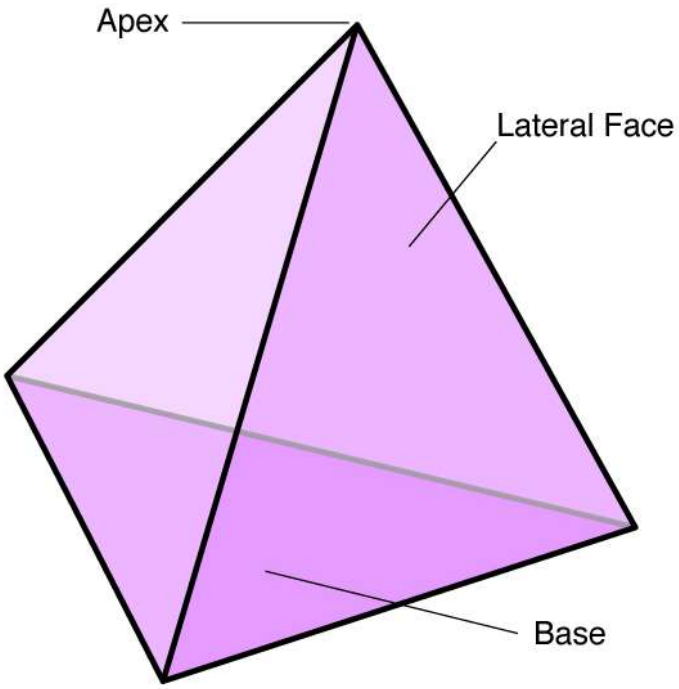
3D Geometry



- Frustum of a cone is a polyhedron that remains when the top off a cone is sliced with a flat cut parallel to its base. The remaining top of a cone is called “truncated cone” or “flat-top cone”.



- Triangular prism is a polyhedron with two parallel, congruent triangular bases connected by three rectangular or parallelogram sides.



- Triangular Pyramid is also known as a Tetrahedron.